

Unit 4 - Applied Commercial and Quality Principles in Engineering.

TECHNICAL EVALUATION REPORT.

An Integrated Study of Quality Management Systems and Value Management Frameworks within Advanced Industrial Automation.

Case Study Platform: The Robopod Standard Workstation.

Host Enterprise: Mechatronic Solutions (Birmingham, UK).

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Quality Systems and Value Management in Engineering.

A Study of Mechatronic Solutions Birmingham and the Robopod Training Workstation.

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Executive Summary.

This engineering report evaluates the operational and commercial performance of Mechatronic Solutions, based in Birmingham, by reviewing its primary automated product: the Robopod training workstation. In modern industrial automation, companies must manage tight profit margins and complex technical designs. This environment requires a precise alignment between formal Quality Management Systems (QMS) and strict Value Management (VM) frameworks. Enterprises that treat quality control and cost engineering as separate tasks often suffer from high defect rates, long lead times, and reduced profitability.

This investigation resolves those issues by conducting a large, data-driven value analysis across the 10 main stages of the Robopod assembly process. The analysis identified a major operational bottleneck during manual electrical wiring. Differences in manual assembly, drawing misinterpretations, and waiting times for parts create a significant cost center, averaging £2,000 per workstation unit. As a result, a detailed process assessment reveals that 45% of this allocation is non-value-added waste that does not improve the machine's final performance or customer satisfaction.

To eliminate this waste, this report proposes a major engineering redesign: replacing manual point-to-point panel wiring with pre-fabricated, keyed, plug-and-play modular wiring harnesses. This engineering modification delivers measurable financial and operational improvements:

- Cuts unit manufacturing costs by £900 per Robopod build.
- Lowers total electrical assembly time from 8.5 hours to 5.0 hours.
- Eradicates internal rework expenses, dropping them from £250 to £0 per unit.

To sustain these efficiencies without creating new production risks, this report shows how to add the modular harness framework directly into the company's ISO 9001 quality infrastructure. Using the Plan-Do-Check-Act (PDCA) continuous improvement cycle, standard work instructions will be rewritten to lock in a "Right First Time" (RFT) philosophy on the shop floor.

In the end, this project proves that value management and quality assurance are highly interdependent. When properly done to work hand-in-hand, they allow Mechatronic Solutions to protect its premium profit margins, shorten customer lead times, and secure a sustainable competitive advantage in the UK automation market.

Introduction.

Macroeconomic Pressures and Applied Commercial Principles.

Engineering enterprises do not operate in a different technical vacuum. Commercial engineering needs balancing advanced design ideas with strict financial limits. In the industrial automation and robotics integration sector, projects are typically won through fixed-price commercial tenders.

When a project budget is fixed by a contract, any unexpected increase in labor hours, material scrap, or component rework directly reduces the organization's gross profit margins. Applied commercial principles require engineering firms to design for manufacturability and assemble for efficiency. Every step on the shop floor must justify its resource consumption by adding measurable performance, reliability, or capability to the final asset.

Also, the fast-paced growth of Industry 4.0 automation systems means engineering firms face clients who demand shorter delivery cycles alongside best equipment reliability. This environment creates high pressure on traditional manufacturing workflows, making older production methods a financial risk.

The Theoretical Intersect of Quality Systems and Value Engineering.

To maintain financial discipline while meeting high technical standards, companies must combine two major terms: Quality Management Systems (QMS) and Value Management (VM).

Quality management provides a structured, documented framework of policies, process responsibilities, and clear operational controls. This system shifts an organization away from reactive defect detection (catching mistakes after production) and toward proactive defect prevention (designing processes so errors cannot happen). This approach ensures that manufactured products meet strict customer requirements and adhere with national engineering standards.

Value Management is a structured, team-based approach used to optimize the value of a product or service. Value is defined mathematically as the optimization of function relative to cost: $\text{value} = \text{function} / \text{cost}$.

Value management does not mean cutting costs by reducing material quality or removing features. Instead, it identifies and removes non-value-added costs, expenses that use up company time, labor, or factory space without adding any value to the final machine's capabilities.

When an engineering firm adds these two systems, it creates a powerful production workflow. Value management removes unnecessary complexity from the design, while quality management ensures the remaining processes are consistently executed without defects.

Operational Bounds and Investigative Scope.

The objective of this technical report is to assess how combining quality and value frameworks produces a strong market advantage for Mechatronic Solutions in Birmingham. The analysis uses the Robopod training workstation as its core product case study.

The first part of this report provides a detailed analysis of international quality standards, specifically comparing the operational benefits of ISO 9001 (Quality Management) and ISO 14001 (Environmental Management). It examines Total Quality Management (TQM) principles, the practical use of the PDCA continuous improvement loop, and specific shop-floor quality control techniques like Statistical Process Control (SPC) and planned maintenance.

The second part of the report uses these theoretical terms through an industrial Value Analysis exercise. This exercise maps out the 10 orderly stages of the Robopod production line. Using real-time manufacturing data, it separates process waste, assesses alternative modular wiring procedures through a decision matrix, and outlines a risk-balanced implementation plan designed to increase company profitability and process capability.

Company Background: Mechatronic Solutions Birmingham. Industrial Automation and Robotics Integration Sector.

The industrial automation and robotics integration sector is a cornerstone of advanced manufacturing. Integrators bridge the gap between high-level robotics manufacturers and factory-floor applications by designing, building, and programming custom systems.

The industry is quickly shifting toward Industry 4.0 paradigms. This change requires integrating physical machinery with data analytics, Internet of Things (IoT) sensors, and cloud networks.

Modern system integrators do not just supply mechanical labor. They deliver smart, adaptable production systems designed to reduce unit cycle times, remove human error, and protect manufacturers from fluctuating labor costs.

Competitive Environment in the Midlands Engineering Hub.

Operating from Kings Norton, Birmingham, Mechatronic Solutions is located at the center of the UK's advanced engineering hub. The Midlands marketplace has a dense network of automation suppliers competing for top-level automotive, aerospace, and medical device contracts.

In this high-risk environment, differentiations depend on an integrator's ability to balance custom engineering with short lead times. Traditional bespoke automation poses financial risks, as long design phases and unexpected commissioning delays can quickly mess-up fixed-price project budgets.

To build a sustainable competitive advantage, local integrators must turn away from high-risk, unstandardized designs. Instead, they need to implement standardized, flexible modular hardware architectures that reduce internal production waste.

Product Overview: The Robopod Workstation.

Design Architecture and Automation Components.

The Robopod is a standardized, plug-and-play robotic enclosure designed to replace single-use, custom automation lines. Built on a rigid steel base, the cell features integrated forklift pockets for rapid factory redeployment.

The system uses an open, reconfigurable architecture. It combines a high-speed 6-axis robotic arm with a central programmable logic controller (PLC) and an interactive Human-Machine Interface (HMI).

The product features a uniform safety circuit, clear guarding, and universal pneumatics. This allows engineers to quickly swap out end-of-arm tooling and component fixtures to support various automated tasks.

Product Lifecycle and Quality Criticality.

The primary value of the Robopod is its reusable lifecycle. When a specific manufacturing contract ends, the platform can be re-tooled and re-programmed for a completely different production use, preserving the client's capital investment.

Because the system works across dynamic, multi-turn lifecycles, manufacturing quality is highly critical. Every internal connection, terminal wire, and structural component must survive continuous industrial cycles.

Any underlying issue introduced during initial assembly will cause operational failures later on. For automated production systems, even a minor internal quality issue can trigger emergency line stoppages for the client, resulting in expensive downtime penalties and damaging the integrator's market reputation.

Part 2: Importance of Quality in Modern Engineering.

Customer Expectations and Global Competition.

In modern industrial automation, quality is no longer a distinct marketing advantage; it is a necessary entry requirement for global supply chains. Advanced manufacturing clients expect high equipment reliability and little component variance.

Integrators compete on a global scale with manufacturers from highly automated economies. To maintain a competitive edge, UK companies must deliver short lead times and zero-defect installations.

If an integrator does not meet these quality baselines, international clients can easily shift their investment to global competitors who use highly standardized, automated assembly lines.

Safety Requirements and Industry Legal Compliance.

Automation systems present serious kinetic and electrical hazards on the factory floor. Industrial units must comply with strict safety regulations, including national electrical standards and machine safety procedures.

Compliance requires using dual-channel safety circuits, interlocking guard switches, and risk assessments.

Not meeting these safety baselines creates immediate legal issues. It can also lead to disastrous equipment breakdowns that put factory operators at risk and result in costly regulatory shutdowns.

Appraisal Matrix: Financial and Operational Consequences of Quality Issues.

Quality failures damage an engineering company's financial health across multiple areas. A comparison grid categorizes these impacts into individual internal and external cost centers.

- **Internal Failure Costs:** Issues caught before client delivery, including structural rework labor, scrapped wiring looms, and repeat testing cycles.
- **External Failure Costs:** Issues discovered after commissioning, including emergency engineer call-outs, transit damage claims, and late-delivery contract penalties.
- **Operational Bottlenecks:** Rework cycles distract senior technicians away from new builds, which stalls the assembly line and reduces overall factory capacity.
- **Reputational Damage:** Repeated field issues erode client trust, disqualifying the company from future automated supply chain tenders.

Quality Management Systems (QMS).

Definition, Core Objectives, and Process-Based Approaches.

A Quality Management System is a structured network of documented policies, operational procedures, and clear responsibilities designed to produce consistent product outcomes.

The core objective is to move an organization away from reactive problem-solving and toward proactive process control.

By laying out production workflows as a series of connected processes, a QMS ensures that inputs, such as raw steel and electronic components, are turned into finished goods using repeatable, measured steps.

Integration of QMS Frameworks within Shop-Floor Operations.

A QMS only adds value when it is deeply integrated into daily shop-floor workflows. This integration relies on standard work instructions, clean visual management boards, and digital assembly logs.

Technicians use these tools to log assembly milestones and trace component batch numbers in real time.

By embedding quality checkpoints directly into the physical assembly line, the organization makes sure that work standards are maintained by operational staff, rather than being treated as a separate administrative exercise.

ISO Standards within Engineering Organisations.

ISO 9001: Quality Management Requirements for Robopod Assembly.

ISO 9001 sets out the international standard for process-driven quality management systems. For a platform like the Robopod, this standard needs strict engineering change control, clear calibration tracking for assembly tools, and robust competency records for technicians.

The framework demands that every manufacturing step follow the approved engineering drawings.

By applying these controls, the company ensures that any engineering changes are updated across all assembly documentation, preventing the use of outdated designs on the factory floor.

As mentioned by David Hoyle in the ISO 9001 Quality Management Systems Manual, true quality assurance demands that an organization turns away from reactive defect detection and instead enforces a documented, process-driven architecture where engineering change controls are effortlessly integrated into daily floor activities.

ISO 14001: Environmental Management and Automation Component Sustainability.

ISO 14001 provides a framework for measuring and reducing an organization's environmental effect. In an automation facility, this standard focuses on lifecycle sustainability, scrap metal segregation, and efficient energy consumption during active machine testing.

It requires teams to design systems that reduce waste, choose recyclable components, and balance logistics to lower carbon emissions.

By adopting these environmental practices, the company reduces waste-disposal costs and fulfills the strict green procurement requirements of major aerospace and automotive clients.

Comparative Matrix: Synergies and Direct Operational Trade-offs.

While ISO 9001 and ISO 14001 focus on different primary objectives, their operational architectures share deep similarities.

- **System Synergies:** Both standards use a process-based structure, rely on regular internal audits, and demand continuous improvement.
- **Documentation Efficiencies:** Organizations can use integrated management systems to run joint audits, which greatly decrease administrative overhead.
- **Operational Trade-offs:** Green material alternatives required by environmental guidelines can sometimes introduce longer procurement lead times.
- **Resource Balances:** Administrative teams must balance the drive for manufacturing speed with the necessary compliance tracking for both standards.

Total Quality Management (TQM).

Origins and Fundamental Pillars of Company-Wide Ownership.

Total Quality Management is a comprehensive management philosophy focused on adding quality across every level of an organization. TQM removes the traditional boundary between production staff and quality inspectors by making every employee responsible for process integrity.

The system relies on three main pillars: an absolute focus on the customer, data-driven decision-making, and continuous process improvement.

Under a TQM framework, every department, from procurement to final delivery, acts as an internal customer, ensuring that quality standards are met at every handoff point.

Employee Empowerment Frameworks versus Operational Limitations.

A core strategy of TQM is empowering assembly technicians to flag and fix quality issues immediately. Line workers have the authority to pause production if they spot an engineering error or a component defect.

While this autonomy catches defects early, it can create operational challenges if it is not supported by a collaborative corporate culture.

Without proper training, frequent line stoppages can disrupt production schedules and create friction between assembly teams and project managers focused on delivery deadlines.

The PDCA Continuous Improvement Cycle.

Theoretical Framework: Plan, Do, Check, Act.

The Plan-Do-Check-Act cycle is a four-step iterative methodology used to manage and improve manufacturing processes.

- Plan: Establish performance baselines, locate process bottlenecks, and design targeted engineering fixes.
- Do: Implement the proposed process changes on a small, controlled trial batch to test effectiveness.
- Check: Gather production data from the trial run and measure the results against original KPIs to verify improvements.
- Act: Standardize successful changes across all production lines by updating official work instructions and training manuals.

Application to Robopod Assembly Line Diagnostics.

When manual wiring errors cause project delays, the PDCA cycle provides a structured way to fix the process. In the Plan phase, engineers audit the assembly

line and discover that unstandardized wiring layouts are causing frequent connection errors.

During the Do phase, the team introduces pre-fabricated modular wiring harnesses on a single trial unit.

In the Check phase, engineers run automated diagnostic tests on the trial unit, confirming a significant drop in continuity faults. Finally, in the Act phase, management updates the official build manual, making modular harnesses the mandatory standard for all future workstations.

Quality Assurance Systems (QA).

Prevention-Driven Quality Planning versus Post-Production Detection.

Quality Assurance focuses on optimizing manufacturing processes to prevent defects from occurring in the first place. This differs fundamentally from Quality Control, which relies on inspecting finished goods to catch mistakes after production.

A prevention-driven approach uses tools like Failure Mode and Effects Analysis (FMEA) to identify and eliminate potential failure points during the design stage.

By focusing on process capability, QA avoids the high costs of scrapping completed assemblies, reducing material waste and optimizing factory throughput.

Audit Infrastructure and Engineering Change Note Control.

Maintaining a reliable production process requires regular internal audits and strict configuration controls. Quality audits verify that assembly teams are following approved work instructions and using calibrated equipment.

When a design needs an update, the team issues a formal Engineering Change Note (ECN).

This control workflow ensures that design updates are reviewed, costed, and signed off by senior engineering figures before being rolled out to the shop floor. This prevents dangerous mix-ups between old and new component versions.

Quality Control Techniques (QC).

Manual and Digital Inspection Procedures.

Quality Control provides the frontline tools needed to check product compliance before final dispatch. Technicians use manual inspections, such as physical weld checks and dimensional measurements, to catch structural defects.

These are paired with digital inspection tools, including laser scanners and digital multi-meters, to verify precision tolerances.

Using both manual and digital verification methods allows the team to catch structural and electronic flaws early, preventing defective systems from reaching the client handover phase.

Frontline quality control relies on a hybrid framework of manual and digital verification procedures to confirm product compliance prior to client handover. Technicians execute manual inspections, including physical weld integrity assessments and physical dimensional measurements of the rigid steel chassis, to catch structural defects. These are paired with digital inspection tools, utilizing laser scanners to check precision tolerances and digital multi-meters to map terminal connectivity, preventing defective sub-assemblies from advancing through the assembly line.

Representative Sampling and Batch Integrity Verification.

For high-volume components like fasteners and structural brackets, inspecting every individual piece is impractical. Instead, teams use representative sampling to check batch quality using statistical baselines.

Quality teams pull a randomized sample from each incoming batch and test it against specific performance tolerances.

If a sample fails to meet those metrics, the entire component batch is rejected and put away. This prevents unverified or sub-standard materials from entering the main assembly line.

For high-volume, standardized components such as structural fasteners, rivets, and terminal brackets, individual inspection is logistically unviable. The quality engineering team applies statistical representative sampling protocols to verify incoming batch integrity. Randomized samples are drawn from every supplier delivery and tested against exact engineering stress-tolerances. If a single sample fails these metrics, the entire batch is quarantined and rejected, ensuring unverified materials never compromise the main production line.

PLC Functional Testing and Simulation Control Protocols.

The core intelligence of an automated system relies on its control software. Quality control for software requires structured PLC functional testing and simulation protocols.

Engineers test the system using digital twin software to simulate physical operations and check for code bugs before the machine is even built.

Once assembled, the physical unit undergoes comprehensive dry-run testing. This process checks sensor responses, safety stop controls, and pneumatic movements against simulated baselines to ensure the control system is completely stable.

The core operational intelligence of the Robopod workstation requires advanced verification. Engineers execute comprehensive PLC functional testing paired with digital-twin simulation protocols. Before physical assembly, control code bugs are isolated and resolved within a simulated virtual environment. Once physical build milestones are achieved, the physical workstation undergoes rigorous dry-run validation cycles to test dual-channel safety circuits, interlocking guard switches, and real-time HMI touch-screen telemetry against simulated baselines.

Condition Monitoring and Planned Torque Tool Maintenance Profiles.

Long-term machine reliability depends on precision assembly and predictive upkeep. Technicians use condition monitoring techniques, such as thermal imaging of active control panels and vibration analysis on robotic joints, to catch assembly variances early.

This is paired with a strict, planned maintenance schedule for factory assembly tools.

Calibrating electric torque drivers every 500 operating hours ensures that structural fasteners are tightened to exact engineering specifications, preventing components from loosening over time due to operational vibrations.

Long-term mechanical reliability is kept through predictive upkeep and strict assembly controls. Technicians utilize active condition monitoring techniques, such as thermal imaging of loaded electrical control enclosures and vibration analysis of the active six-axis robotic arm, to capture microscopic process shifts. This is structurally reinforced by a planned maintenance profile for shop-floor assembly tools, mandating the digital calibration of electric torque drivers every five hundred operating hours to guarantee all fasteners are secured to exact engineering specifications, completely eliminating vibration-induced loosening over the platform's multi-turn lifecycle.

Statistical Process Control (SPC): X-bar and Range Variance Tracking.

Statistical Process Control uses data charts to monitor and control process stability. Quality teams track critical dimensions, such as frame squareness and door alignments, across production runs using X-bar and Range charts.

These charts record the mean values and range variations of measurements over time, mapping them against defined upper and lower control limits.

If a measurement trends toward a control limit, it flags a process shift or tool wear before an actual defect occurs. This allows production teams to adjust machinery proactively and keep the process stable.

The Right First Time (RFT) Philosophy.

Interaction with Lean Engineering Principles.

The Right First Time philosophy focuses on eliminating waste by ensuring every assembly step is executed perfectly on the first attempt. This aligns directly with lean engineering principles, which target the removal of process waste, over-processing, and unnecessary motion.

RFT treats rework not just as a minor inconvenience, but as a major operational failure that wastes labor and capacity.

By building quality directly into the assembly process, an organization can maintain a smooth production flow, shorten manufacturing lead times, and lower production costs.

Financial Benefits of Poka-Yoke Error-Proofing Structures.

Implementing the RFT philosophy on the shop floor relies on using physical and digital poka-yoke (error-proofing) mechanisms. These mechanisms include keyed electrical connectors, asymmetric structural brackets, and smart tool sensors that prevent assembly if parts are misaligned.

These error-proofing designs prevent assembly mistakes from happening, eliminating the need for expensive teardowns and manual rework.

Saving hours of troubleshooting labor directly improves product profit margins, helps protect project delivery dates, and strengthens the company's financial position.

Part 3: Value Management Principles.

Conceptual Definitions: Maximising Function relative to Total Cost.

Value Management is a structured, function-oriented methodology used to improve

the overall worth of an engineering asset. Worth is defined by the mathematical relationship between the essential functions a product performs and the total lifecycle cost required to deliver those functions.

The primary objective is to increase this ratio. This is achieved either by enhancing the product's performance capabilities at a stable cost center, or by reducing production costs without reducing reliability.

This discipline differs from aggressive cost-cutting. Traditional cost-cutting focuses on reducing material specifications, which often reduces product quality.

Value management preserves or improves all core functional performance requirements while identifying and removing unnecessary manufacturing costs.

This aligns completely with the foundational methodology established by Lawrence Miles in Techniques of Value Analysis and Engineering, which says that product value must be enhanced by targeting and eliminating non-value-added costs, which are expenses that consume labor or factory space without adding to the product's essential functional capabilities.

Eliminating Non-Value-Added Waste from Manufacturing Operations.

To balance production costs, operations are grouped into value-added and non-value-added activities. Value-added steps directly alter the form, fit, or function of raw materials to meet customer specifications.

Non-value-added activities consume company time, labor, and factory floor space without adding any value to the final product.

These non-value-added steps represent process waste. Examples include moving components multiple times, waiting for engineering designs, and repairing manual errors.

Identifying and removing these non-value-added steps streamlines production and lowers manufacturing costs without affecting structural quality.

Value Management Tools and Techniques.

Function Analysis System Technique (FAST) and Root-Cause Mapping.

Value management relies on structured diagnostic tools, specifically Function Analysis System Technique (FAST) mapping and Ishikawa root-cause diagrams. FAST mapping translates a physical product into distinct, verb-noun functional requirements.

This approach forces engineering teams to focus on what a component does rather than what it is. For example, an electrical enclosure's function is expressed as "house electronics" and "protect operators."

When a function generates high costs, teams use Ishikawa root-cause diagrams and the "Five Whys" technique to trace inefficiencies back to their origin.

Combining functional analysis with root-cause tracking allows engineers to pinpoint exactly which designs or processes are creating non-value-added costs.

Competitive Advantage Through Quality and Value.

Strategic Synergy: High Quality Performance and Low Cost Bases.

The combination of rigorous quality control and disciplined value management creates a strong competitive advantage. Organizations that implement only one of these frameworks often face major problems.

A company focused solely on quality may build defect-free products that are too expensive for the market, while a company focused only on cost reduction risks delivering low-cost items with high failure rates.

Combining these systems creates a powerful production workflow. Value management simplifies the product design and removes process waste, while quality management ensures the remaining processes are consistently executed without defects.

This synergy allows Mechatronic Solutions to underbid competitors while protecting its premium profit margins.

Within the Midlands engineering hub, advanced automation projects are predominantly secured through rigid, fixed-price commercial tenders. In this economic environment, an engineering firm cannot establish a sustainable competitive advantage by treating quality control and cost engineering as isolated tasks. By strategically pairing disciplined value management with a robust quality management system, Mechatronic Solutions establishes a real head-start market advantage over regional rivals.

First, the company achieves cost leadership through waste cut. Traditional cost-cutting strategies compromise material specifications, introducing severe failure risks. In contrast, applying Lawrence Miles' value analysis principles allows the enterprise to mathematically optimize the function-to-cost ratio. Cutting down the forty-five percent non-value-added manual wiring waste reduces the total Robopod build cost by six percent, or nine hundred pounds per unit. This allows the company to aggressively underbid competitors during fixed-price bidding phases while safely protecting its gross profit margins.

On top of that, the company creates strong quality stand-outs and market speed. While low-cost rivals risk delivering systems with high field-failure rates, adding the modular harness design directly into the ISO 9001 framework guarantees zero-defect installations. Shifting the electrical assembly duration from eight and a half hours to five hours sharply cuts manufacturing lead times. Delivering highly reliable, Industry 4.0-ready assets faster than international competitors fulfills the strict procurement requirements of tier-one aerospace and automotive clients, securely positioning Mechatronic Solutions as a premium, low-risk systems integrator.

Value Analysis (VA) Exercise Framework.

Exercise Objectives, Boundaries, and 10-Stage Chronological Scope.

This value analysis exercise assesses the standard manufacturing cycle of the Robopod workstation. The primary aim is to reduce total production costs by 15% while maintaining all performance and safety requirements.

The scope of the analysis is bounded by a 10-stage orderly set lifecycle, tracking the product from initial requirements gathering to final delivery.

This clear boundary ensures that all engineering, material, assembly, and quality verification costs are accurately recorded. This data allows the team to pinpoint exactly where non-value-added waste is impacting the production line.

Information Phase.

Establish Cost Baseline Across the 10-Stage Robopod Production Cycle.

The information phase shows a clear cost baseline across the entire manufacturing cycle. The target cost for a standard Robopod workstation is set at £15,000.

Production data and timesheets were collected to distribute labor and material costs across the 10 distinct stages of the build.

- Stage 1: Customer Requirements Gathering - Labor: £300, Materials: £0, Total: £300
- Stage 2: Design and CAD Modelling - Labor: £1,200, Materials: £0, Total: £1,200
- Stage 3: Component Procurement - Labor: £200, Materials: £0, Total: £200
- Stage 4: Frame Fabrication - Labor: £800, Materials: £2,500, Total: £3,300
- Stage 5: Mechanical Assembly - Labor: £1,500, Materials: £4,000, Total: £5,500

- Stage 6: Electrical Wiring - Labor: £1,400, Materials: £600, Total: £2,000
- Stage 7: PLC Programming - Labor: £1,000, Materials: £0, Total: £1,000
- Stage 8: Functional Testing - Labor: £600, Materials: £0, Total: £600
- Stage 9: Quality Inspection - Labor: £400, Materials: £0, Total: £400
- Stage 10: Packaging and Delivery - Labor: £200, Materials: £300, Total: £500
- Master Financial Baseline Summary - Total Labor: £7,600, Total Materials: £7,400, **Total Project Cost: £15,000**

Analysis Phase.

Classification of Functional Value-Added (VA) versus Non-Value-Added (NVA) Steps.

The analysis phase reviews each manufacturing stage to separate value-added functions from non-value-added waste.

Stages 1, 2, 4, 5, 7, and 8 are classified as pure Value-Added (VA) steps because they directly build, program, or verify the product features that the customer is paying for.

In contrast, Stage 6 (Electrical Wiring) is identified as a major source of inefficiency. While routing power is an essential function, auditing this stage reveals that 45% (£900) of its £2,000 cost is non-value-added waste.

Timesheet data shows that technicians spend hours troubleshooting loose terminals, correcting manual wiring layout errors, and waiting for missing components on the shop floor.

Creative Phase.

Idea Generation for Eradicating Manual Wiring Bottlenecks.

The creative phase focuses on brainstorming alternative design methods to remove the manual wiring bottleneck identified in Stage 6. The engineering team produced three distinct structural concepts.

- Concept A: Modular Plug-and-Play Wiring Harnesses - Replace manual point-to-point panel wiring with pre-fabricated, keyed wiring looms that plug directly into integrated heavy-duty connectors.
- Concept B: Outsource Complete Enclosure Panels - Outsource the assembly, wiring, and testing of all electrical control enclosures to a specialized external panel builder.

- Concept C: Augmented Reality Guided Assembly - Provide shop-floor technicians with AR headsets that project real-time, 3D digital wiring paths directly onto the physical chassis.

Evaluation Phase.

Feasibility, Financial Benefit, and Implementation Risk Decision Matrix.

The evaluation phase uses a structured decision system to grade the three proposed concepts. Each option is scored from 1 (Poor) to 10 (Excellent) across four critical engineering metrics.

- Concept A (Modular Harness) - Technical Feasibility: 9, Upfront Cost: 8, Risk Management: 9, Cost Savings: 8, Total Score: 34/40 (Selected)
- Concept B (Panel Outsourcing) - Technical Feasibility: 6, Upfront Cost: 4, Risk Management: 5, Cost Savings: 7, Total Score: 22/40
- Concept C (AR Guided Assembly) - Technical Feasibility: 4, Upfront Cost: 2, Risk Management: 3, Cost Savings: 9, Total Score: 18/40

Concept A was selected as the optimal solution. It offers excellent technical productivity and less deployment risk.

While Concept C promises high long-term savings, its high upfront software costs and technical complexity make it unviable for immediate rollout. Concept B reduces internal manufacturing control, creating unacceptable risks around external component quality and supply chain delays.

Development and Reporting Phase.

The Plug-and-Play Modular Wiring Harness Plan and Projected Financial Return.

The development phase shows the implementation plan for the selected plug-and-play modular wiring harness. The engineering department will update the Robopod's electrical schematics to transition away from individual, manual wire terminations.

The enclosure will be updated to use standardized, heavy-duty multi-pin connectors, allowing the main wiring harness to be manufactured as a single plug-in kit.

This engineering change alters the cost structure of Stage 6. Manual labor costs **drop from £1,400 to just £200**, as connection times are cut significantly. Material procurement costs for the pre-made harness kit **rise from £600 to £900**.

This results in **a new total stage cost of £1,100**, down from the original £2,000 baseline. This modification produces **a direct saving of £900 per Robopod unit**, reducing total manufacturing costs **by 6%** and successfully removing non-value-added assembly waste.

Part 4: Evaluation of Findings.

Critical Interrelationship Between Value Modifications and QMS Integrity.

Value modifications and Quality Management Systems exist in a deeply dependent relationship. Modifying a product's design to reduce non-value-added costs cannot succeed if it is treated as an isolated exercise. For Mechatronic Solutions, transitioning to a plug-and-play modular wiring harness is an output of value management. However, this engineering change cannot function without immediate, corresponding updates to the ISO 9001 quality infrastructure.

If the physical wiring harness is improved but the official quality assurance documentation, assembly drawings, and inspection checklists are not updated under strict Engineering Change Note control, a severe process gap opens. Technicians on the shop floor would be left checking a modular assembly using outdated point-to-point testing protocols.

True operational success occurs when value management simplifies the physical process, which in turn elevates the capability of the QMS. By using physical poka-yoke features on the new modular connectors, the business designs out the potential for human error. This design adjustment mainly reduces the complexity of floor inspections, reduces the burden on internal auditors, and makes sure that the "Right First Time" manufacturing target is structurally locked into the company's daily operations.

Potential Cultural Barriers to Shop-Floor Changes.

Introducing new engineering workflows often creates different cultural disputes within established manufacturing environments. Shop-floor technicians who possess years of specialized experience in manual point-to-point panel wiring may view the introduction of pre-fabricated, plug-and-play harnesses as a threat to their technical skill set or job security. This friction can lead to passive resistance, such as staff intentionally ignoring new work instructions, rushing through new inspection steps, or blaming the new components for unrelated production delays.

To overcome these cultural barriers, management must avoid imposing the change through purely top-down mandates. Instead, the changes must be managed using Total Quality Management principles that empower floor staff. Assembly technicians

should be involved early during the prototype testing phase of the new wiring harness.

By positioning the change as a way to remove frustrating, non-value-added troubleshooting tasks rather than an attempt to de-skill their roles, management can build collective ownership. This approach completely changes potential shop-floor critics into proactive supporters of the new manufacturing process.

Prioritised Recommendations and Implementation Strategy.

Priority Mapping via Cost-Benefit Structuring.

To execute these changes effectively without messing-up ongoing factory operations, project recommendations are grouped using a structured cost-benefit system. This approach separates immediate operational improvements from longer-term strategic investments.

- Priority 1: High Benefit, Low Cost (Immediate Rollout): Transition the Robopod workstation line to the plug-and-play modular wiring harness. Simultaneously, update all relevant ISO 9001 standard work instructions and engineer-approved schematics. This step delivers immediate financial savings of £900 per unit with minimal upfront tooling expense.
- Priority 2: High Benefit, High Cost (Systemic Integration): Establish a comprehensive Statistical Process Control program to track structural frame squareness. Integrate ISO 9001 quality audits and ISO 14001 environmental audits into a single administrative workflow to reduce corporate overheads.
- Priority 3: Low Benefit, High Cost (Deferred): Introduce augmented reality guided assembly headsets. While technically advanced, the high software licensing costs and extensive training requirements mean this option should be deferred until Robopod manufacturing volumes scale up significantly.
- Priority 4: Low Benefit, Low Cost (Routine Maintenance): Enhance local scrap segregation protocols for aluminum alloy offcuts within the frame fabrication area to maintain baseline compliance with ISO 14001 guidelines.

Strategic Risk Mitigation and Procurement Diversification Plan.

Moving from internal manual assembly to outsourced pre-fabricated components changes the company's operational risk profile. While it removes shop-floor assembly errors, it makes Mechatronic Solutions dependent on external suppliers for critical wiring components. A delay from a single wiring vendor could stall the entire Robopod assembly line, creating expensive project delivery bottlenecks.

To manage this supply chain risk, the procurement department must implement a strict diversification strategy:

- Dual Sourcing: Avoid single-source supply agreements. The specialized wiring harness must be qualified with two independent UK-based manufacturers to guarantee material availability.
- Rigorous Supplier Audits: Potential wiring vendors must pass formal quality capability audits, verifying their adherence to ISO 9001 standards before receiving production contracts.
- Buffer Stock Strategy: Maintain a rolling inventory buffer of two complete wiring harness kits at the Birmingham facility to insulate production from temporary transport delays or supplier backlogs.

Key Performance Indicators.

Progress must be checked regularly using quantifiable performance metrics. The following key performance indicators will be tracked monthly by the quality engineering team to measure the success of the new processes against historical production baselines.

- Electrical Wiring Defect Rate
 - Historical Baseline: 5.0% internal error rate during final testing.
 - Post-Implementation Target: < 0.5% (Six Sigma quality target).
 - Measurement Method: Automated diagnostic continuity logs.
- Total Electrical Assembly Duration
 - Historical Baseline: 8.5 manufacturing hours per unit.
 - Post-Implementation Target: 5.0 manufacturing hours per unit.
 - Measurement Method: Digital shop-floor timesheet tracking.
- Rework Unit Expense
 - Historical Baseline: £250 average rework cost per workstation build.
 - Post-Implementation Target: £0 per workstation build (Complete elimination).
 - Measurement Method: Materials scrap logs and corrective labor tracking.
- Supplier On-Time In-Full (OTIF) Delivery
 - Historical Baseline: 92.0% compliance from raw component vendors.
 - Post-Implementation Target: > 98.5% compliance from certified harness partners.
 - Measurement Method: Inbound logistics receipt audits.

Conclusion.

This technical investigation shows that combining structured Quality Management Systems with disciplined Value Management principles is needed for maintaining commercial profitability. Relying on older manufacturing workflows creates significant business risks, such as long production hours and high rework costs on the shop floor.

Assessing the Robopod assembly line identified a clear operational bottleneck in Stage 6 (Electrical Wiring), where manual connection errors and technical misinterpretations accounted for a major source of process waste.

Replacing manual point-to-point panel wiring with pre-fabricated, keyed plug-and-play modular harnesses resolves this inefficiency. This engineering modification produces a direct cost saving of £900 per Robopod build, while significantly shortening assembly times and removing internal rework expenses.

To ensure these improvements are sustainable, the new modular architecture must be formally added into the company's ISO 9001 quality framework using the Plan-Do-Check-Act cycle. Also, this approach shows that value management and quality control work best as an integrated system. By continually removing process waste and standardizing work instructions, Mechatronic Solutions can protect its operating margins, shorten client lead times, and maintain a strong competitive advantage within the UK automation sector.

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Part 5: Appendices.

Appendix A: Chronological Robopod 10-Stage Process Flow Diagram.

This section outlines the process sequence for assembling the Robopod workstation.

- Stage 1: Customer Requirements Gathering: Technical consultation to map out the client's automation footprint, safety zones, and specific production environment.

- Stage 2: Design and CAD Modelling: Engineers generate the master solid-model assemblies, pneumatic schematics, and structural sheet-metal drawings.
- Stage 3: Component Procurement: The ERP system automatically orders the raw steel, fixings, specialized sensors, and the core 6-axis robotic arm from approved suppliers.
- Stage 4: Frame Fabrication: Structural technicians cut, weld, and grind the primary heavy steel chassis before sending it out for industrial powder coating.
- Stage 5: Mechanical Assembly: Team builds the physical unit by bolting down the floor plates, mounting safety guarding panels, and installing the 6-axis robot arm.
- Stage 6: Electrical Wiring (The Process Bottleneck): Technicians manually run point-to-point power cables, sensor lines, and communication links inside the main control enclosure.
- Stage 7: PLC Programming: Systems engineers upload the operational code, configure the safety system parameters, and calibrate the HMI touch screens.
- Stage 8: Functional Testing: Run extensive dry-run validation cycles to make sure pneumatic actuators, safety switches, and logic sub-routines perform as needed.
- Stage 9: Quality Inspection: Final quality review to make sure the build matches the engineering drawing dimensions and check compliance with national safety standards.
- Stage 10: Packaging and Delivery: The completed Robopod unit is packaged, secured to a heavy-duty transport pallet, and loaded onto flatbed transit for customer delivery.

Appendix B: Value Stream Map of Electrical Assembly Bottlenecks.

The following timeline details the operational waste found during the original Stage 6 manual wiring process.

- Step 1: Layout Control Enclosure (Value-Added)
 - Duration: 1.5 Hours.
 - Observation: Technician physically positions the primary electrical trunking, din-rail mountings, and terminal blocks inside the chassis.
- Step 2: Drawing Discrepancy Queries (Non-Value-Added Waste)
 - Duration: 1.8 Hours.
 - Observation: Production stalls while the assembly team waits for engineering to clarify schematic layout changes and correct manual wire numbering errors.
- Step 3: Route Power and Signal Cables (Value-Added)
 - Duration: 1.0 Hour.

- Observation: Core task of running and pulling the physical power cables and low-voltage sensor cables through the containment trunking.
- Step 4: Component Location Delays (Non-Value-Added Waste)
 - Duration: 0.5 Hours.
 - Observation: Production stops as the technician leaves the assembly area to search the warehouse for missing cable glands and correct wire markings.
- Step 5: Cut, Strip, and Crimp Terminations (Value-Added)
 - Duration: 1.2 Hours.
 - Observation: Technician manually strips back individual wire ends and crimps on ferrules to prepare the lines for final connection.
- Step 6: Fault Rectification and Rework (Non-Value-Added Waste)
 - Duration: 1.5 Hours.
 - Observation: Technician reconstructs sections of the panel to fix loose wire terminations and reversed polarity errors caught during preliminary testing.
- Step 7: Final Terminal Connection (Value-Added)
 - Duration: 1.0 Hour.
 - Observation: The remaining valid circuits are wired into the main PLC modules and checked for solid terminal retention.

Appendix C: Full Cost Allocation and Target Savings Excel Sheet.

This section expands on the cost parameters used for the value analysis exercise.

- Target Cost Reduction Limit: 15.0% of the overall build cost (£2,250 target savings threshold).

Detailed Stage-by-Stage Cost Breakdown Matrix.

Stage	Process Activity Descriptors	Baseline Labor Cost (£)	Baseline Material Cost (£)	Master Cost Baseline (£)	Value Classification	Revised Labor Cost (£)	Revised Material Cost (£)	Projected Cost Outcome (£)	Net Savings Achieved (£)
1	Customer Requirements	£300	£0	£300	Value-Added	£300	£0	£300	£0

2	Design & CAD Modelling	£1,200	£0	£1,200	Value-Added	£1,200	£0	£1,200	£0
3	Component Procurement	£200	£0	£200	Value-Added	£200	£0	£200	£0
4	Frame Fabrication	£800	£2,500	£3,300	Value-Added	£800	£2,500	£3,300	£0
5	Mechanical Assembly	£1,500	£4,000	£5,500	Value-Added	£1,500	£4,000	£5,500	£0
6	Electrical Wiring	£1,400	£600	£2,000	45% NVA Waste	£200	£900	£1,100	£900
7	PLC Programming	£1,000	£0	£1,000	Value-Added	£1,000	£0	£1,000	£0
8	Functional Testing	£600	£0	£600	Value-Added	£600	£0	£600	£0
9	Quality Inspection	£400	£0	£400	Value-Added	£400	£0	£400	£0
10	Packaging & Delivery	£200	£300	£500	Value-Added	£200	£300	£500	£0
-	Total Project Summary	£7,600	£7,400	£15,000	-	£6,400	£7,700	£14,100	£900

Appendix D: Quality System Inspection and Poka-Yoke Work Instructions.

Mechatronic Solutions Technical Work Instruction Document: MS-WI-E06

- System Applicability: Robopod Standard Workstation Platforms.
- Operational Task Focus: Installation of Main Control Modular Wiring Harnesses.
- Revision Level: 2.0 (Updated post Value Analysis Review).

1. Mandatory Pre-Assembly Safety Protocol.

Before starting work, technicians must make sure the assembly area is clear of hazards. Ensure the main workstation isolation switch is locked off, tag-out warning tags are attached to the power distribution box, and proper anti-static wrist straps are grounded to the workstation frame.

2. Visual Component Inspection and Setup.

Examine the incoming pre-fabricated wiring harness box. Verify the internal quality-release stamp matches the supplier's batch number. Check the harness insulation for cuts or deep crimping marks, and ensure the multi-pin connector housings are free of debris or cracked pins.

3. Execution of Keyed Poka-Yoke Insertion Steps.

- Align the Connector: Position the heavy-duty harness plug in front of the main enclosure socket plate. Locate the machined, asymmetric alignment keyway tab on the outer edge of the connector housing.
- Insert the Component: Push the harness connector into the socket. The asymmetric keyway acts as a physical error-proofing tool, preventing insertion if the plug is rotated incorrectly. If you meet structural resistance, do not use force. Re-align the indexing slots and try again.
- Secure the Position: Flip the dual integrated steel locking latches over the housing pins. Press down until they click into place. This lock creates a dust-tight seal and ensures individual terminal contacts cannot work loose over time due to machine vibrations.

4. Post-Assembly Testing and Quality Log Submission.

Connect the digital multi-meter interface harness to the auxiliary testing panel on the workstation. Run the automated diagnostic test file on the console to check continuity across all circuits.

Once the test returns a zero-fault status, log the continuity data file into the factory ERP terminal to complete the step under ISO 9001 guidelines before moving the machine to Stage 7.

